

Practical-Week5

1. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>
#include <string.h>

#define BUFFER_SIZE 100

int main() {
    int pipefd[2];
    pid_t pid;

    char buffer[BUFFER_SIZE];
    int data[] = {10, 20, 30, 40, 50};
    int n = 5;

    struct timespec start, end;

    // Create anonymous pipe
    if (pipe(pipefd) == -1) {
        perror("pipe");
        exit(EXIT_FAILURE);
    }

    // Start measuring time
    clock_gettime(CLOCK_MONOTONIC, &start);

    // Create child process
    pid = fork();

    if (pid < 0) {
        perror("fork");
        exit(EXIT_FAILURE);
    }
```

```
pawan_kumar@LAPTOP-SSD0PUQ2:~$ nano producer_consumer.c
pawan_kumar@LAPTOP-SSD0PUQ2:~$ gcc producer_consumer.c -o producer_consumer
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ./producer_consumer
Producer (Parent): Sending data...
Produced: 10

Consumer (Child): Receiving data...
Produced: 20
Produced: 30
Produced: 40
Produced: 50
Consumed: 10
Consumed: 20
Consumed: 30
Consumed: 40
Consumed: 50

Communication completed.
Communication time: 0.00065114 seconds
Data transferred: 20 bytes
Communication efficiency (throughput): 30715.46 bytes/sec
pawan_kumar@LAPTOP-SSD0PUQ2:~$ |
```

```
pawan_kumar@LAPTOP-SSD0PUQ2:~$ strace -f -e trace=process,ipc,read,write,close,wait4 ./producer_consumer
execve("./producer_consumer", ["/producer_consumer"], 0x7ffc47b2a7a8 /* 26 vars */) = 0
close(3) = 0
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"... , 832) = 832
close(3) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x7b4c5d0d6a10) = 883
[pid 882] close(3) = 0
[pid 883] close(4) = 0
[pid 882] write(1, "Producer (Parent): Sending data...., 35Producer (Parent): Sending data...
) = 35
[pid 883] write(1, "\n", 1 <unfinished ...>

[pid 882] write(4, "\n\0\0\0", 4 <unfinished ...>
[pid 883] <... write resumed> = 1
[pid 882] <... write resumed> = 4
[pid 883] write(1, "Consumer (Child): Receiving data"... , 36 <unfinished ...>
Consumer (Child): Receiving data...
[pid 882] write(1, "Produced: 10\n", 13 <unfinished ...>
Produced: 10
[pid 883] <... write resumed> = 36
[pid 882] <... write resumed> = 13
[pid 883] read(3 <unfinished ...>
[pid 882] write(4, "\24\0\0\0", 4 <unfinished ...>
[pid 883] <... read resumed>, "\n\0\0\0", 4) = 4
[pid 882] <... write resumed> = 4
[pid 883] write(1, "Consumed: 10\n", 13 <unfinished ...>
Consumed: 10
[pid 882] write(1, "Produced: 20\n", 13 <unfinished ...>
Produced: 20
[pid 883] <... write resumed> = 13
[pid 882] <... write resumed> = 13
[pid 883] read(3 <unfinished ...>
[pid 882] write(4, "\36\0\0\0", 4 <unfinished ...>
[pid 883] <... read resumed>, "\24\0\0\0", 4) = 4
[pid 882] <... write resumed> = 4
[pid 883] write(1, "Consumed: 20\n", 13 <unfinished ...>
Consumed: 20
[pid 882] write(1, "Produced: 30\n", 13 <unfinished ...>
Produced: 30
[pid 883] <... write resumed> = 13
[pid 882] <... write resumed> = 13
[pid 883] read(3 <unfinished ...>
[pid 882] write(4, "(\0\0\0", 4 <unfinished ...>
[pid 883] <... read resumed>, "\36\0\0\0", 4) = 4
[pid 882] <... write resumed> = 4
[pid 883] write(1, "Consumed: 30\n", 13 <unfinished ...>
Consumed: 30
```

```

[pid 883] read(3 <unfinished ...>
[pid 882] write(4, "\0\0\0", 4 <unfinished ...>
[pid 883] <... read resumed>, "\36\0\0\0", 4) = 4
[pid 882] <... write resumed> = 4
[pid 883] write(1, "Consumed: 30\n", 13 <unfinished ...>
Consumed: 30
[pid 882] write(1, "Produced: 40\n", 13 <unfinished ...>
Produced: 40
[pid 883] <... write resumed> = 13
[pid 882] <... write resumed> = 13
[pid 883] read(3 <unfinished ...>
[pid 882] write(4, "2\0\0\0", 4 <unfinished ...>
[pid 883] <... read resumed>, "\0\0\0\0", 4) = 4
[pid 882] <... write resumed> = 4
[pid 883] write(1, "Consumed: 40\n", 13 <unfinished ...>
Consumed: 40
[pid 882] write(1, "Produced: 50\n", 13 <unfinished ...>
Produced: 50
[pid 883] <... write resumed> = 13
[pid 882] <... write resumed> = 13
[pid 883] read(3 <unfinished ...>
[pid 882] close(4 <unfinished ...>
[pid 883] <... read resumed>, "2\0\0\0", 4) = 4
[pid 882] <... close resumed> = 0
[pid 883] write(1, "Consumed: 50\n", 13 <unfinished ...>
Consumed: 50
[pid 882] wait4(-1 <unfinished ...>
[pid 883] <... write resumed> = 13
[pid 883] close(3) = 0
[pid 883] exit_group(0) = ?
[pid 883] +++ exited with 0 +++
<... wait4 resumed>, NULL, 0, NULL) = 883
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=883, si_uid=1000, si_status=0, si_utime=0, si_stime=0} ---
write(1, "\n", 1) = 1
)
write(1, "Communication completed.\n", 25Communication completed.
) = 25
write(1, "Communication time: 0.00803399 s...", 39Communication time: 0.00803399 seconds
) = 39
write(1, "Data transferred: 20 bytes\n", 27Data transferred: 20 bytes
) = 27
write(1, "Communication efficiency (throug...", 57Communication efficiency (throughput): 2489.42 bytes/sec
) = 57
exit_group(0) = ?
+++ exited with 0 +++
pawan_kumar@LAPTOP-SSD0PUQ2:~$ |

```

Report: The strace command is used to trace system calls made by the producer and consumer processes, such as read(), write(), close(), and wait(). The output shows that the parent produced 10, 20, 30, 40, 50 and the child consumed the same data successfully, with a communication time of about 0.00037 seconds and throughput of 54233.17 bytes/sec.

2. Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using fork(), pipe(), dup2(), and exec() system calls.01:32 PM

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int pipefd[2];
    pid_t pid1, pid2;

    if (pipe(pipefd) == -1) {
        perror("pipe");
        exit(1);
    }

    // First child: executes ls -l
    pid1 = fork();

    if (pid1 == -1) {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0) {
        // Redirect stdout to pipe
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp ls");
        exit(1);
    }

    // Second child: executes grep ".c"
    pid2 = fork();
```

```

pawan_kumar@LAPTOP-SSD0PUQ2:~$ gcc ls_grep.c -o ls_grep
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ./ls_grep
-rw-r--r-- 1 pawan_kumar pawan_kumar 0 Aug 5 18:03 Hello.java
-rw-r--r-- 1 pawan_kumar pawan_kumar 0 Aug 5 17:59 HelloCopy.java
-rw-r--r-- 1 pawan_kumar pawan_kumar 0 Aug 5 17:56 Main.java
drwxr-xr-x 12 pawan_kumar pawan_kumar 4096 Aug 12 16:39 OSSP
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16344 Aug 25 18:59 builtin
-rw-r--r-- 1 pawan_kumar pawan_kumar 1044 Aug 25 18:59 builtin.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 40 Aug 17 05:05 destination.txt
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16128 Aug 17 05:05 filecopy
-rw-r--r-- 1 pawan_kumar pawan_kumar 498 Aug 17 05:04 filecopy.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 15952 Aug 12 05:43 hello
-rw-r--r-- 1 pawan_kumar pawan_kumar 78 Aug 12 05:43 hello.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 73 Aug 29 08:18 info.txt
-rwxr-xr-x 1 pawan_kumar pawan_kumar 15960 Sep 18 13:47 ls_grep
-rw-r--r-- 1 pawan_kumar pawan_kumar 103 Sep 18 13:47 ls_grep.c
prw-r--r-- 1 pawan_kumar pawan_kumar 0 Aug 29 08:22 myfifo
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16384 Aug 25 18:53 path_search
-rw-r--r-- 1 pawan_kumar pawan_kumar 1039 Aug 25 18:53 path_search.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 52 Aug 12 04:52 practice.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 52 Aug 6 03:21 practice.txtMangoOrangeBananaAppleGuavaPapayaCherryApple
-rw-r--r-- 1 pawan_kumar pawan_kumar 12 Aug 6 03:18 practice2.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 53 Aug 7 06:25 practice2.txt.save
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16344 Aug 25 18:39 process
-rw-r--r-- 1 pawan_kumar pawan_kumar 1641 Aug 25 18:39 process.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16216 Aug 17 05:18 process_state
-rw-r--r-- 1 pawan_kumar pawan_kumar 937 Aug 17 05:18 process_state.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16400 Sep 18 13:35 producer_consumer
-rw-r--r-- 1 pawan_kumar pawan_kumar 1696 Sep 18 13:35 producer_consumer.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 0 Aug 5 17:56 program.py
-rw-r--r-- 1 pawan_kumar pawan_kumar 25 Aug 17 05:12 sample.txt
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16552 Aug 17 14:24 shell
-rw-r--r-- 1 pawan_kumar pawan_kumar 1141 Aug 17 14:24 shell.c
drwxr-xr-x 2 pawan_kumar pawan_kumar 4096 Aug 17 05:48 shell_project
-rw-r--r-- 1 pawan_kumar pawan_kumar 40 Aug 17 05:04 source.txt
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16080 Aug 12 15:50 task2
-rw-r--r-- 1 pawan_kumar pawan_kumar 441 Aug 12 15:49 task2.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16272 Aug 25 18:57 variable_expand
-rw-r--r-- 1 pawan_kumar pawan_kumar 690 Aug 25 18:56 variable_expand.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16304 Aug 25 18:50 waitpid_demo
-rw-r--r-- 1 pawan_kumar pawan_kumar 945 Aug 25 18:50 waitpid_demo.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16208 Aug 25 18:44 zombie
-rw-r--r-- 1 pawan_kumar pawan_kumar 627 Aug 25 18:44 zombie.c
pawan_kumar@LAPTOP-SSD0PUQ2:~$

```

```

pawan_kumar@LAPTOP-SSD0PUQ2:~$ nano ls_grep.c
pawan_kumar@LAPTOP-SSD0PUQ2:~$ gcc ls_grep.c -o ls_grep
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ./ls_grep
-rw-r--r-- 1 pawan_kumar pawan_kumar 1044 Aug 25 18:59 builtin.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16128 Aug 17 05:05 filecopy
-rw-r--r-- 1 pawan_kumar pawan_kumar 498 Aug 17 05:04 filecopy.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 78 Aug 12 05:43 hello.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 774 Sep 18 13:51 ls_grep.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16384 Aug 25 18:53 path_search
-rw-r--r-- 1 pawan_kumar pawan_kumar 1039 Aug 25 18:53 path_search.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 52 Aug 12 04:52 practice.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 52 Aug 6 03:21 practice.txtMangoOrangeBananaAppleGuavaPapayaCherryApple
-rw-r--r-- 1 pawan_kumar pawan_kumar 12 Aug 6 03:18 practice2.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 53 Aug 7 06:25 practice2.txt.save
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16344 Aug 25 18:39 process
-rw-r--r-- 1 pawan_kumar pawan_kumar 1641 Aug 25 18:39 process.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16216 Aug 17 05:18 process_state
-rw-r--r-- 1 pawan_kumar pawan_kumar 937 Aug 17 05:18 process_state.c
-rwxr-xr-x 1 pawan_kumar pawan_kumar 16400 Sep 18 13:35 producer_consumer
-rw-r--r-- 1 pawan_kumar pawan_kumar 1696 Sep 18 13:35 producer_consumer.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 1141 Aug 17 14:24 shell.c
drwxr-xr-x 2 pawan_kumar pawan_kumar 4096 Aug 17 05:48 shell_project
-rw-r--r-- 1 pawan_kumar pawan_kumar 40 Aug 17 05:04 source.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 441 Aug 12 15:49 task2.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 4911 Sep 18 13:49 trace.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 690 Aug 25 18:56 variable_expand.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 945 Aug 25 18:50 waitpid_demo.c
-rw-r--r-- 1 pawan_kumar pawan_kumar 627 Aug 25 18:44 zombie.c
pawan_kumar@LAPTOP-SSD0PUQ2:~$ strace -f -e trace=process,ipc,dup2,execve,close,wait4 ./ls_grep
execve("./ls_grep", ["/ls_grep"], 0x7ffe07655c38 /* 26 vars */) = 0
close(3) = 0
close(3) = 0
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0x759a1fbbca10) = 1280
[pid 1279] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD <unfinished ...>
[pid 1280] close(3) = 0
[pid 1280] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD <unfinished ...>
[pid 1279] <... clone resumed>, child_tidptr=0x759a1fbbca10) = 1281
[pid 1280] dup2(4, 1) = 1
[pid 1279] close(3) = 0
[pid 1280] close(4 <unfinished ...>
[pid 1279] close(4) = 0
[pid 1280] <... close resumed>
[pid 1279] wait4(1280 <unfinished ...>
[pid 1281] close(4 <unfinished ...>
[pid 1280] execve("/usr/local/sbin/ls", ["ls", "-l"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1281] <... close resumed> = 0
[pid 1280] <... execve resumed> = -1 ENOENT (No such file or directory)
[pid 1281] dup2(3, 0 <unfinished ...>
[pid 1280] execve("/usr/local/bin/ls", ["ls", "-l"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1281] <... dup2 resumed> = 0
[pid 1280] <... execve resumed> = -1 ENOENT (No such file or directory)
[pid 1281] close(3 <unfinished ...>
[pid 1280] execve("/usr/sbin/ls", ["ls", "-l"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1281] <... close resumed> = 0
[pid 1280] <... execve resumed> = -1 ENOENT (No such file or directory)
[pid 1281] execve("/usr/local/sbin/grep", ["grep", ".c"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1280] execve("/usr/bin/ls", ["ls", "-l"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1281] <... execve resumed> = -1 ENOENT (No such file or directory)
[pid 1280] execve("/usr/local/bin/grep", ["grep", ".c"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1281] execve("/usr/sbin/grep", ["grep", ".c"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1280] <... execve resumed> = 0
[pid 1281] <... execve resumed> = -1 ENOENT (No such file or directory)
[pid 1280] execve("/usr/bin/grep", ["grep", ".c"], 0x7fff90ffe538 /* 26 vars */ <unfinished ...>
[pid 1281] close(3) = 0
[pid 1280] close(3) = 0
[pid 1281] close(3 <unfinished ...>

```

